Measuring Interdisciplinarity: A Multi-Component Indicator Panel for Research Evaluation

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2026

Outline

The Problem

- Interdisciplinary research (IDR) is widely promoted by policy initiatives worldwide (National Academies, 2005).
- Yet **measurement remains unsettled**:
  - Wang & Schneider (2020): 23 measures, low mutual correlations — "no single indicator can unequivocally identify" IDR.
  - Leydesdorff et al. (2019): Rao-Stirling dominated by disparity, limited discriminatory power.
  - Cantone (2024): IDR is a "polysemous construct" — multiple meanings, no single number suffices.
- **Fundamental mismatch**: multidimensional phenomenon $\longleftrightarrow$ scalar indicators.

This Paper's Contributions

1. **Taxonomy**: four conceptual dimensions $\times$ four methodological families.
2. **Panel**: three-component indicator triple (Δ, S, E) spanning diversity, coherence, and cross-field effect.
3. **Demonstration**: toy data + department-level case study show the panel uniquely discriminates integrators, polymaths, and specialists.
4. **Robustness**: analytical proof that discrimination holds under similarity-matrix perturbation.

Stirling's Diversity Framework

Stirling (2007): diversity requires three properties:

Variety How many categories are represented.

Balance How evenly distributed across categories.

Disparity How different the categories are from each other.

Shannon entropy and Herfindahl capture only variety + balance.

Rao-Stirling index:

$$\Delta = \sum_{i \neq j} d_{ij}\, p_i\, p_j$$

- p_i: proportion in category i
- $d_{ij} = 1 - s_{ij}$: dissimilarity
- Incorporates all three properties

Historical Evolution & Typology

- OECD (1998), Morillo et al. (2003) — tripartite typology:
  - **Multidisciplinary** Juxtaposes perspectives, no synthesis.
  - **Interdisciplinary** Integrates data, methods, theories — "greater than the sum of its parts" (Wagner et al., 2011).
  - **Transdisciplinary** Transcends disciplinary epistemologies.
- Aksnes et al. (2026): 42% of papers rated as *both* multi- and interdisciplinary — not a clean partition but a spectrum.
- Directionality matters: input-side vs. output-side IDR (Zhou et al., 2023).

The Polysemy Problem

- Cantone (2024): IDR is a **polysemous construct —** physicist + biologist, data scientist, social theorist all called "interdisciplinary."
- Three measurement *approaches* that cluster differently:
  - **Cognitive**: diversity of references, methods, frameworks.
  - **Grouping/Social**: co-authorship, affiliations.
  - **Semantic**: keywords, abstracts, full-text analysis.
- Self-reported $\neq$ bibliometric IDR (Zwanenburg, 2022; Aksnes et al., 2026: $r \approx 0.15$).
- "Big" vs. "small" interdisciplinarity (Morillo et al., 2003).

Epistemological & Validity Challenges

- **Classification instability**: WoS 254 categories achieve only $\sim$50% alignment with citation-based clusters (Boyack, 2005).
- **Horizontal disciplines**: Nature, Science, PNAS classified as "Multidisciplinary Sciences" — low IDR scores despite spanning the disciplinary spectrum.
- **Validity crisis** (Zwanenburg et al., 2022):
  - Content validity: indicators capture only 1–2 of 3 diversity properties.
  - Domain validity: ambiguous journal-to-category mappings.
  - Coherence validity: notionally equivalent measures diverge.
- Rafols (2019): responsible metrics need both analytical validity *and* social robustness.

Taxonomy Overview

Dimension	Reference-based	Citation-based	Text-based	Network-based
Diversity	Rao-Stirling, Simpson, Shannon, Gini, Hill	—	Semantic similarity	—
Coherence	Bibliographic coupling density	—	—	Betweenness, clustering
Diffusion	—	Cross-field cites, citing diversity	—	—
Novelty	—	—	—	Recombination metrics

- Reference-based/diversity cell is heavily populated; others sparse.
- Coherence and diffusion are orthogonal to diversity (Wang & Schneider, 2020).
- No existing indicator spans multiple dimensions $\Rightarrow$ need a multi-component panel.

Diversity Indicators: Key Results

- **Variety + Balance only**: Shannon H, Simpson D_{Sim}, Brillouin, Gini — intercorrelate $r \in [0.60, 1.00]$, but only weakly with disparity measures.
- **Rao-Stirling** $\Delta = \sum_{i \neq j} d_{ij}\, p_i\, p_j$: most widely used, but highly sensitive to s_{ij} specification.
- Wang & Schneider (2020): 8 RS variants; pairwise r as low as 0.18.
- **Hill-type true diversity** (Zhang et al., 2016):

$$ {}^2D^S = \frac{1}{\sum_{i,j} s_{ij}\, p_i\, p_j} = \frac{1}{1 - \Delta} $$

  Interpretation as "effective number of disciplines"; satisfies replication principle.
- **Similarity matrix**: construction (Salton vs. Ochiai cosine) is a first-order determinant of results.

Validity Assessment (Zwanenburg et al., 2022)

21 measures evaluated against 8 criteria:

1. **Applicability**: (a) Multi-object, (b) Size independence.
2. **Integration evidence**: measure should reflect actual knowledge integration.
3. **Discipline identification**: (a) Valid allocation, (b) Completeness, (c) Low classification bias.
4. **Diversity capture**: (a) Sensitivity to variety + balance + disparity, (b) Decomposability.

- Only 6 of 21 met criterion 2 (integration evidence).
- **No single measure** satisfied all 8 criteria.
- $\sim$30% of references assigned to multiple WoS categories $\Rightarrow$ allocation ambiguity cascades into inflated scores.

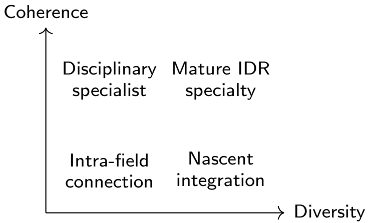
Coherence Indicators

Key idea (Rafols & Meyer, 2009): do diverse inputs form a unified intellectual fabric, or are they merely juxtaposed? **Mean linkage strength**:

$$S = \frac{2}{N(N-1)} \sum_{a<b} s_{ab}$$

where $s_{ab} = \cos(\mathbf{r}_a, \mathbf{r}_b)$ via bibliographic coupling.

Diversity–Coherence matrix:



- Empirically orthogonal to diversity ($r < 0.5$; Wang & Schneider, 2020).
- Captures depth of integration, not breadth.

Diffusion Indicators

- Look **forward**: who cites the work across fields?
- Replace reference vector $\mathbf{p}(x)$ with citation vector $\mathbf{q}(x)$.
- Leydesdorff et al. (2019) — **DIV indicator**:

$$\mathrm{DIV}_c = \frac{n_c}{N} \left(1 - G_c\right) \frac{\sum_{i \neq j} d_{ij}}{n_c(n_c - 1)}$$

  monotonically increasing in each component.
- Highest-scoring journals: *PLOS ONE*, *Science*, *Nature*.
- Xiang et al. (2025): knowledge-base IDR $\uparrow$ acceptance; topic IDR $\downarrow$ acceptance.
- **Conceptual tension**: diffusion is an *outcome*, not a property of the research process (Cantone, 2024).

Novelty Indicators

- Is a combination **atypical**, not just diverse?
- **Uzzi et al. (2013)**: z-score against permuted citation networks.
- **Wang et al. (2017)**: timed novelty — weight by recency of first combination.
- **Validation failures**:
  - Bornmann (2019): no correlation with F1000Prime expert assessments.
  - Fontana et al. (2020): Uzzi measure correlates with Rao-Stirling diversity instead.

$\Rightarrow$ **Excluded from our panel**: high data requirements + failed external validation.

- Novelty remains an important *conceptual* dimension; awaits better operationalization.

Panel Definition: (Δ, S, E)

Diversity Δ Rao-Stirling index over aggregate reference proportions:

$$\Delta = 1 - \sum_{i,j} s_{ij}\, p_i\, p_j$$

Coherence S Mean bibliographic-coupling cosine across publication pairs:

$$S = \frac{1}{\binom{n}{2}} \sum_{k<l} \cos(\mathbf{r}_k, \mathbf{r}_l)$$

Cross-field effect E Fraction of citations from outside each publication's primary category:

$$E = \frac{\text{cross-field citations}}{\text{total citations}}$$

Each component captures an **independent** dimension; only Δ depends on the similarity matrix $\mathbf{S}$.

Toy Data: Setup

Five categories; three researcher archetypes:

Similarity matrix:

$$\mathbf{S} = \begin{pmatrix} 1.00 & 0.60 & 0.40 & 0.10 & 0.30 \\ 0.60 & 1.00 & 0.50 & 0.15 & 0.20 \\ 0.40 & 0.50 & 1.00 & 0.35 & 0.10 \\ 0.10 & 0.15 & 0.35 & 1.00 & 0.05 \\ 0.30 & 0.20 & 0.10 & 0.05 & 1.00 \end{pmatrix}$$

Researcher profiles:

- **A** (Integrator): each paper references multiple distant categories.
  $\mathbf{p}_A = (0.30, 0.08, 0.18, 0.35, 0.10)$
- **B** (Polymath): 5 single-field papers, one per category.
  $\mathbf{p}_B = (0.20, 0.20, 0.20, 0.20, 0.20)$
- **C** (Specialist): all papers in condensed matter + materials.
  $\mathbf{p}_C = (0.64, 0.27, 0.06, 0.00, 0.03)$

Toy Data: Results — Single Indicators Fail

Researcher	Δ	S	E	**Type**
A (integrator)	0.559	0.733	0.600	Cross-disciplinary
B (polymath)	0.580	0.000	0.063	Polymathic breadth
C (specialist)	0.245	0.881	0.211	Disciplinary

Key finding: $\Delta_A \approx \Delta_B$ (0.559 vs. 0.580).

- Diversity *alone* cannot distinguish integrator from polymath.
- S reveals the difference: $S_A = 0.73$ (shared references) vs. $S_B = 0.00$ (orthogonal publications).
- E confirms at impact level: $E_A = 0.60$ vs. $E_B = 0.06$.
- **Only the full triple uniquely characterizes each type.**

Sensitivity Analysis: Robustness Guarantees

Proposition (uniform perturbation $s_{ij} \to s_{ij} + \varepsilon$):

$$\Delta_{\mathrm{new}} = \Delta_{\mathrm{old}} - \varepsilon\,(1 - H) \qquad \text{where } H = \textstyle\sum_i p_i^2.$$

Corollary: ranking inverts at critical perturbation

$$\varepsilon^* = -\frac{\Delta_A - \Delta_B}{H_A - H_B} = 0.351$$

— requires shifting *every* similarity by $>$35%.

- Non-uniform perturbation experiments confirm robustness under realistic scenarios.
- **Crucially**: S and E are *exactly invariant* under any similarity-matrix perturbation.
- $\Rightarrow$ Panel discrimination is substantially more robust than any single-indicator approach.

Interpretation Framework: Pattern Taxonomy

Pattern	Δ	S	E	**Interpretation**
Genuine integrator	High	High	High	Cross-disciplinary synthesis
Polymath	High	Low	Low	Breadth without integration
Specialist	Low	High	Low	Narrow disciplinary focus
Emergent / sparse data	Low	Low	Any	Defer to expert review

Illustrative thresholds for operational deployment:

Classification	Δ	S	E
Genuine integrator	≥ 0.40	≥ 0.30	≥ 0.30
Polymath (non-integrative)	≥ 0.40	< 0.15	< 0.15
Specialist (reclassify)	< 0.35	any	any
Ambiguous (expert review)	else	else	else

Edge Cases and Failure Modes

1. **Breadth-without-depth**: High Δ without integration.
   Mitigation: require S and E thresholds.
2. **Non-standard publication penalty**: IDR journals often have lower impact factors.
   Mitigation: field-normalized citation indicators.
3. **Incommensurable citation norms**: citation rates differ across fields by an order of magnitude.
   Mitigation: normalize E by field baselines.
4. **Misclassification persistence**: specialist stays on IDR track.
   Mitigation: automatic reclassification trigger ($\Delta < 0.35$).
5. **Early-career data sparsity**: < 15 publications $\Rightarrow$ unstable estimates.
   Mitigation: minimum publication threshold + confidence intervals.
6. **Gaming via strategic co-authorship**: inflates Δ without integration.
   Mitigation: cross-check against S.

Practical Implementation Pipeline

Cantone (2024) — five-step pipeline:

1. **Unit of analysis**: paper, author, institution, journal.
2. **Taxonomy**: WoS (254 categories), OECD (40), OpenAlex.
3. **Classification method**: journal-based, reference-based, semantic (LLM), collaboration-based.
4. **Operational definition**: dimension + formula selection.
5. **Aggregation**: elementary (per-paper mean) vs. collective (pooled references).

Key finding: measures cluster by method, not by dimension (Cantone, 2024).
Implication: document every methodological choice explicitly.

Institutional & National Deployment

Institutional self-assessment:

- Δ and S computable from internal data.
- E requires external citation data (WoS, Scopus, OpenAlex).
- Two-component profile (Δ, S) still discriminates integrators from polymaths.

National agencies:

- Spain's CNEAI — **Campo 0**: first dedicated IDR evaluation track (2023).
- Criteria map onto panel:
  - "Interdisciplinar" $\leftrightarrow$ high Δ + high S.
  - "Multidisciplinar" $\leftrightarrow$ high variety $N \geq 2$.
- Transdisciplinary: absent from criteria — no standard bibliometric indicator exists.

Open Problems (I)

1. **Self-reported vs. bibliometric IDR**: correlations only $r \approx 0.15$ (Aksnes et al., 2026). Researchers assess IDR via collaboration and methods, not reference patterns.
2. **LLM-based similarity matrices**: Cantone (2025) shows partial agreement with citation-based estimates; could democratize access to disparity-sensitive IDR measurement.
3. **Uncertainty quantification**: Nakhoda et al. (2023) — bootstrap 95% CIs span up to 0.6 for Rao-Stirling. Point estimates can be misleading.
4. **IDR and quality**: short-term citation penalty for IDR (Lariviere & Gingras, 2010), but longer horizons show greater lasting impact. Direction matters: knowledge-base IDR $\uparrow$ acceptance, topic IDR $\downarrow$ (Xiang et al., 2025).

Open Problems (II)

5. **Distribution-based approaches**: Zhou et al. (2023) IKF framework (B, I, H) decomposes what Rao-Stirling aggregates.
6. **Participatory indicators**: Marres & de Rijcke (2020) — “engaging indicators” that combine scientometrics with stakeholder workshops. Challenge: scaling to national level.
7. **Transdisciplinary measurement**: no standard bibliometric indicator captures engagement with non-academic partners (Borlaug & Svartefoss, 2025). Spain’s Campo 0 omits it entirely.

Department-Level Case Study: Setup

Hypothetical Physics & Materials Science department, 7 researchers.
Six WoS categories (C_1–C_6): condensed matter, materials science, physical chemistry, optics, electrical engineering, nanoscience.

Researcher	**Stage**	**Pubs**	$\mathbf{p} = (p_1, \ldots, p_6)$	**Cites**	**Cross-field**
Chen	S	42	(0.35, 0.30, 0.20, 0.05, 0.05, 0.05)	520	35%
Al-Rahman	M	35	(0.20, 0.20, 0.20, 0.20, 0.15, 0.05)	280	11%
Kowalski	E	18	(0.65, 0.25, 0.06, 0.02, 0.01, 0.01)	85	12%
Nguyen	S	55	(0.25, 0.25, 0.15, 0.10, 0.15, 0.10)	890	47%
Romero	M	28	(0.10, 0.15, 0.05, 0.05, 0.10, 0.55)	195	72%
Karlsson	E	12	(0.50, 0.30, 0.10, 0.05, 0.03, 0.02)	45	18%
Osei	S	48	(0.70, 0.20, 0.05, 0.03, 0.01, 0.01)	680	11%

Case Study: Panel Results

Researcher	Δ	S	E	**Classification**
Chen	0.390	0.42	0.35	Moderate integrator
Al-Rahman	0.610	0.04	0.11	Polymath
Kowalski	0.190	0.53	0.12	Specialist
Nguyen	0.464	0.50	0.47	Strong integrator
Romero	0.374	0.50	0.72	Niche-bridge specialist
Karlsson	0.244	0.58	0.18	Early-career specialist
Osei	0.176	0.58	0.11	Specialist

- **Al-Rahman**: highest Δ in department, but $S \approx 0$ and $E = 0.11 \Rightarrow$ polymath, *not* integrator.
- **Romero**: moderate Δ but exceptional $E = 0.72 \Rightarrow$ bridge-specialist requiring expert review.
- **Nguyen**: exceeds all three thresholds $\Rightarrow$ genuine integrator.

Case Study: Single Indicators Compared

Rank	Shannon H	Rao-Stirling Δ	Cross-field E
1	Al-Rahman (2.43)	Al-Rahman (0.610)	Romero (0.72)
2	Nguyen (2.32)	Nguyen (0.464)	Nguyen (0.47)
3	Chen (2.05)	Chen (0.390)	Chen (0.35)

- Each single indicator produces a **different** "most interdisciplinary" researcher.
- Shannon and RS both rank the polymath #1.
- Cross-field effect ranks the bridge specialist #1.
- **Full panel avoids all three errors**: Al-Rahman's zero coherence unmasks the polymath; Romero is flagged for expert review; Nguyen is confirmed as genuine integrator.

Conclusions

1. **No single indicator is adequate** for measuring IDR. The literature is heavily concentrated on diversity; coherence, diffusion, and novelty are underexplored.
2. The **three-component panel** (Δ, S, E):
   - Spans three orthogonal dimensions.
   - Uniquely discriminates integrators, polymaths, and specialists.
   - Robust under similarity-matrix perturbation ($\varepsilon^* = 0.35$).
   - S and E are exactly invariant to the similarity matrix.
3. **Practical implications**:
   - Evaluation agencies should use multidimensional profiles, not composite scores.
   - Thresholds must be calibrated to local context.
   - Report confidence intervals + methodological specifications.
4. **Next steps**: empirical validation on real university data; integration of LLM-based similarity; uncertainty quantification.

Thank you

Questions?

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